

Table 1: Pass@K Evaluation Results on Code Generation Benchmarks

Models: Meta-LLaMA MTP 7B (4-head). Training: LoRA (rank=64) on CodeContests.  
Pure = Pretrained MTP, Baseline = MTP + SFT, Verifiable = MTP + WMTP (GAE-weighted)

| Dataset      | Problems | Samples | Eval Method             | Temp | Model      | Pass@1        | Pass@5        | Pass@10       | Pass@20       |
|--------------|----------|---------|-------------------------|------|------------|---------------|---------------|---------------|---------------|
| HumanEval    | 80       | 20      | Function execution      | 0.8  | Pure       | 2.94%         | 13.17%        | 23.30%        | 37.50%        |
|              |          |         |                         |      | Baseline   | 2.56%         | 11.85%        | 21.45%        | 35.00%        |
|              |          |         |                         |      | Verifiable | <b>3.25%</b>  | <b>14.27%</b> | <b>24.79%</b> | <b>40.00%</b> |
| MBPP         | 100      | 20      | Assert execution        | 0.2  | Pure       | 17.00%        | 24.58%        | 27.99%        | 31.00%        |
|              |          |         |                         |      | Baseline   | 17.50%        | 23.55%        | 26.28%        | 30.00%        |
|              |          |         |                         |      | Verifiable | <b>17.80%</b> | <b>26.26%</b> | <b>30.29%</b> | <b>35.00%</b> |
| GSM8K        | 100      | 20      | Exact match             | 0.2  | Pure       | 2.95%         | 7.34%         | 10.40%        | 14.00%        |
|              |          |         |                         |      | Baseline   | <b>3.20%</b>  | <b>8.80%</b>  | <b>11.34%</b> | <b>13.00%</b> |
|              |          |         |                         |      | Verifiable | 3.00%         | 8.45%         | 12.16%        | 17.00%        |
| CodeContests | 20       | 20      | stdin/stdout (10 tests) | 0.2  | Pure       | 0.25%         | 1.25%         | 2.50%         | 5.00%         |
|              |          |         |                         |      | Baseline   | <b>1.00%</b>  | <b>4.25%</b>  | <b>6.97%</b>  | <b>10.00%</b> |
|              |          |         |                         |      | Verifiable | 0.50%         | 2.50%         | 5.00%         | 10.00%        |

Notes: Pass@K computed using unbiased estimator (Chen et al., 2021). Temperature selected via search (0.2, 0.8). Best results per dataset in bold.

Pure (Pretrained MTP)

Baseline (MTP + SFT)

Verifiable (MTP + WMTP)